2025 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW) | 979-8-3315-1484-6/25/$31.00 ©2025 IEEE | DOI: 10.1109/VRW66409.2025.00224

An Overview of the 1st International Workshop on Spatial Memory in XR: The Future of Memory Capture and Replay Through XR and AI (XRMemory)

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Figure 1: The concept image of spatial memory in XR where people are experiencing their memories through Augmented/Mixed/Virtual Reality wearable devices. (This image was generated using OpenAI's DALL·E through ChatGPT assistance.)

Abstract

The 1st International Workshop on Spatial Memory in eXtended Reality (XRMemory) aims to explore and advance technologies that enable immersive, memory-rich experiences, allowing users to revisit moments as if they were truly back in those spaces. While photos and videos have traditionally served this purpose, advancements in Augmented Reality (AR), Virtual Reality (VR), and spatial computing now enable far more realistic and immersive memory recall. This workshop investigates how AI and XR technologies can be harnessed to record, generate, and recall user experiences in transformative ways. By enabling individuals to relive their own memories or immerse themselves in others' experiences as if physically present, it seeks to open new avenues for shared connections that transcend the limitations of time and space.

Index Terms: Spatial Memory, XR, AI, AR, VR.

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1 Overview

The goal of this workshop is to explore technologies enabling immersive, memory-rich experiences, allowing users to revisit moments as if they were truly back in those spaces. While traditional media such as photos and videos have long served as tools for recalling past experiences [3], the increasing accessibility of Augmented/Virtual Reality (AR/VR) Head Mounted Displays (HMDs) and the progress in spatial computing now open doors to far more immersive and realistic forms of memory recall across time and space [1, 2]. This workshop aims to bring together experts across Artificial Intelligence (AI), eXtended Reality (XR), AR/VR, graphics, computer vision, and large language models (LLMs) to investigate the technological foundations for capturing, recreating, and sharing spatial experiences. By fostering discussions on how these interdisciplinary technologies can enable richer, more vivid forms of memory retention and experience sharing, we hope to reshape the way of reminding memory across the limits of space and time.

2 Call for Papers

2.1 Topics of Interest

We invite a diverse range of topics, encompassing various disciplines and perspectives including but not limited to:

- AI-driven techniques for enhancing spatial memory recall in the XR environment
- Methods for capturing and reconstructing personal experiences in immersive spaces
- Multisensory integration and its impact on memory retention in virtual settings
- Virtual environment design for ecologically valid spatial memory research

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DOI 10.1109/VRW66409.2025.00224

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- Cognitive and emotional factors influencing memory recall in AR/VR.

2.2 Submission Types

XRMemory workshop welcomes a range of paper types (up to 6 pages, excluding references), including:

- **Position Papers** that share unique insights, opinions, and forward-thinking ideas
- **Research Papers** that present the latest advancements, backed by results, including applications and systems related to spatial memory in XR
- **Survey Papers** that provide a comprehensive overview of the current research landscape, highlighting research gaps and future opportunities
- **Project Papers** that describe the approach and objectives of ongoing or planned projects.

3 Organization

The XRMemory workshop organization consists of eight organizers and two international program committees. Dooyoung Kim, the primary organizer, has collaborated with seven co-organizers from KAIST, NYU, and the University of Calgary who are conducting research related to spatial memory to design this workshop. Additionally, Ken Perlin and Steven Feiner, as members of the international program committee, provide overall feedback on the workshop and deliver keynote presentations.

3.1 Organizers

- Dooyoung Kim (dooyoung.kim@kaist.ac.kr), Senior Researcher (PhD) at KAIST, focuses on AR, spatial computing, and remote collaboration.
- Woontack Woo (wwoo@kaist.ac.kr), Professor at KAIST, specializes in ubiquitous VR/AR and leads the Post-Metaverse Research Center.
- Kangsoo Kim (kangsoo.kim@ucalgary.ca), Assistant Professor at the University of Calgary, researches immersive XR systems to enhance human abilities and social presence.
- Cláudio Silva (csilva@nyu.edu), Professor at NYU, works on data science, computer graphics, and visualization with applications in urban science, sports analytics, and paleontology.
- Dishita Turakhia (d.turakhia@nyu.edu), Faculty at NYU Tandon, designs interactive systems for physical skill learning using XR and generative AI.
- Qi Sun (qisun@nyu.edu), Assistant Professor at NYU, focuses on optimizing XR displays and enhancing audio-visual spatial perception.
- Zhu Wang (zhu.wang@nyu.edu), Postdoctoral Associate at NYU Future Reality Lab (FRL), explores immersive experiences and natural interactions in XR.
- Keru Wang (keru.wang@nyu.edu), PhD Researcher at NYU FRL, enhances collaborative experiences in XR and generative VR authoring.

3.2 International Program Committee

- Ken Perlin (ken.perlin@gmail.com), Professor at NYU, Founder of FRL, known for Perlin Noise, multi-user XR research, and interactive storytelling.
- Steven Feiner (feiner@cs.columbia.edu), Professor at Columbia University, Director of the Computer Graphics and User Interfaces Lab, pioneering work in AR, HCI, and wearable computing.

4 Schedule and Paper Overview

The 1st XRMemory workshop is organized as a half-day in-person workshop at IEEE VR 2025. The tentative schedule is shown in Tab. 1. This workshop brings together innovative research that explores the intersection of memory, attention, and immersive technologies, highlighting how virtual and extended realities can reshape cognitive and social experiences. The six accepted papers present diverse perspectives, including studies on imperceptible environmental transitions in VR and their impact on spatial memory and perception, as well as the preservation of autobiographical memory through telepresence systems. These contributions examine both the theoretical foundations and practical applications of immersive technologies, providing insights into designing adaptive, emotionally connective, and interactive experiences.

Table 1: Workshop schedule

Time	Content
8:20 - 8:40	Registration & Introduction
8:40 - 8:50	Opening Remarks
8:50 - 9:20	Keynote 1 (Ken Perlin)
9:20 - 9:50	Keynote 2 (Steven Feiner)
9:50 - 10:10	Keynote Discussion
10:10 - 10:30	Break
10:30 - 11:40	Paper Session
11:40 - 12:20	Panel & Open Discussion
12:20 - 12:30	Closing

The program expands on tangible and speculative innovations in XR, such as embedding XR memories onto physical objects with "Tangible Moments," context dependency in memory recognition, and leveraging AI and personal data in XR to foster introspection and prospection. Additionally, a novel framework, "Avagent," introduces AI-powered avatars for asynchronous communication in VR/AR, enabling context-rich, interactive, and emotionally nuanced engagement across time and space. Together, these presentations illuminate the transformative potential of immersive technologies for memory, communication, learning, and self-reflection, setting the stage for future advancements and interdisciplinary collaboration.

5 Conclusion

In conclusion, the XRMemory: 1st International Workshop on Spatial Memory in XR aims to lay the groundwork for advancing the field of spatial memory in immersive environments. This workshop fosters collaborative exploration of cutting-edge technologies and methodologies for capturing, replaying, and sharing memory-rich experiences by bringing together experts and innovators across diverse disciplines. We believe this initiative will serve as a pivotal step toward shaping the future of spatial memory in XR, driving impactful research that transcends the boundaries of time and space.

Acknowledgments

The authors thank all authors who submitted and presented their research, the IEEE VR 2025 Workshop, Publication, and General Chairs, as well as the audience for their participation and contributions to the workshop's success.

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